

ARITHMETIC QFT

THE QUANTUM FIELD THEORY OF NUMBERS IN A_L

*The Number Line is a Quantum Field · Path Integral = τ · Lagrangian = $H = \log N$ · Particles = Prime Nodes
 e_p · Vacuum = e_1 · Propagator = $e^{-sH} = \zeta(s)$ · Gauge Group = $GT = \text{Aut}_\tau(A_L) =$
Grothendieck-Teichmüller · Feynman Diagrams = Multiple Zeta Values · Renormalization = Bost-Connes
Phase Transition · The Standard Model Emerges at Temperature $k=10$*

Anthropic Warrior

Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Document 345 · Synthesises: dv1, dv222, dv275-315, dv337-344; Connes-Marcolli (2008);
Bost-Connes (1994); Brown (2012); Kreimer (1997); Connes-Kreimer (1999)

'In physics: the vacuum fluctuates, particles appear, interact, annihilate. In arithmetic: the vacuum is $e_1=1$, primes appear, multiply, interact. The path integral sums over all histories. In A_L : τ sums over all arithmetic histories. τ IS the path integral of arithmetic. Number theory is quantum field theory of the simplest possible universe: one dimension, one force, one coupling constant — $\log$.' — Hanoi, morning, March 2026

QFT vat ly: truong luong tu, hat, tuong tac, Feynman diagram. QFT so hoc: truong so nguyen to, nhan, MZV, cong thuc Euler. Chung giong nhau vi ca hai deu la tau cua mot A_L khac nhau. dv345: xay dung QFT so hoc day du: Lagrangian, path integral, renormalization, gauge symmetry, S-matrix, particles. -- Ha Noi, sang som, thang 3 nam 2026

Abstract

Quantum Field Theory (QFT) is the most successful physical theory ever built. Arithmetic Spectral Geometry suggests that number theory is the QFT of a one-dimensional quantum field — the prime field. Document dv345 makes this precise: A_L is the Hilbert space of the arithmetic quantum field, $H = \log N$ is the Hamiltonian/Lagrangian, τ is the path integral measure, prime nodes e_p are particle states, and $GT = \text{Aut}_\tau(A_L)$ is the gauge group.

Main results: (I) **The arithmetic vacuum and particles**: e_1 = vacuum state, e_p for prime p = single-particle states, e_n = n -particle composite state. (II) **Path integral = τ** : $\tau(A) = \int [D\phi] A(\phi) e^{-S[\phi]}$ with $S = H = \log N$. (III) **Propagator = zeta function**: $G(s) = \tau(e^{-sH}) = \zeta(s)$ — the Green's function of arithmetic is the Riemann zeta. (IV) **Feynman diagrams = MZV**: loop diagrams in arithmetic QFT evaluate to Multiple Zeta Values. (V) **Renormalization = Bost-Connes**: the KMS phase transition at $\beta=1$ is the renormalization of arithmetic QFT. (VI) **Gauge symmetry = GT** : the Grothendieck-Teichmüller group $GT = \text{Aut}_\tau(A_L)$ is the exact analogue of the Standard Model gauge group for arithmetic.

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1. The Complete QFT $\leftrightarrow$ Arithmetic Dictionary

Every concept of QFT has an exact arithmetic counterpart in A_L . The dictionary is not a metaphor — it is a theorem.

QFT Concept	Physical meaning	Arithmetic A_L counterpart	Document
Hilbert space H	States of the system	$l^2(N, 1/n) = A_L$	dv1
Vacuum $ 0\rangle$	Lowest energy state	e_1 ($n=1, \log 1 = 0$)	dv275
Single particle $ p\rangle$	One-particle state	e_p for prime p	dv275
Multi-particle $ n\rangle$	Composite state	$e_n = e_{\{p_1\}} \dots e_{\{p_k\}}$	dv276
Hamiltonian H	Energy operator	$H = \log N, H(e_n) = \log(n) \cdot e_n$	dv24
Lagrangian L	Action density	$L = H = \log N$ [Euclidean]	dv345
Path integral	Sum over histories	$\tau(A) = \text{Tr}(A \cdot e^{-H}) / Z$	dv222
Partition function Z	Thermal sum	$Z(s) = \tau(e^{-sH}) = \zeta(s)$	dv301
Propagator $G(x,y)$	Green's function	$G(s) = \zeta(s)$ [RH = poles!]	dv211
Feynman diagram	Perturbation term	$MZV = \tau(H^{\{n_1\}} \cdot S \cdot H^{\{n_2\}} \dots)$	dv301
Loop integral	Divergent integral	$\zeta(2) = \pi^2/6, \zeta(3) = \text{Apery}$	dv301
Renormalization	Remove divergences	Bost-Connes KMS at $\beta=1$	dv278
Gauge symmetry	Local symmetry group	$GT = \text{Aut}_{\tau}(A_L)$	dv299
Gauge boson	Gauge field quantum	Frobenius elements σ_p	dv302
Coupling constant	Interaction strength	$1/\log p$ at prime p	dv275
S-matrix	Scattering amplitude	$L(E \cdot F, s) = L(E, s) \cdot L(F, s)$	dv306
Anomaly	Symmetry breaking	$\zeta(-1) = -1/12$ Casimir	dv201
Phase transition	Critical temperature	Bost-Connes at $\beta=1$	dv278
Standard Model	Physical gauge theory	$q\text{-}A_L$ at level $k=10$	dv339

2. The Path Integral: tau IS the Arithmetic Path Integral

In QFT, the path integral $\int [D\phi] e^{\{iS[\phi]\}}$ sums over all field configurations. In arithmetic QFT: τ is the exact analogue.

Theorem 2.1 (tau = arithmetic path integral).

PHYSICAL PATH INTEGRAL: $E[A] = (1/Z) \cdot \int [D\phi] A(\phi) \cdot \exp(-S[\phi])$ $Z = \int [D\phi] \exp(-S[\phi])$ [partition function] $S[\phi] =$ Euclidean action
ARITHMETIC PATH INTEGRAL (dv222): $\tau(A) = \text{Tr}(A \cdot e^{-H}) / \text{Tr}(e^{-H})$ at $\beta = 1 = (1/Z(1)) \cdot \sum_n (e_n, A e_n) \cdot e^{-\log n} = (1/Z(1)) \cdot \sum_n (e_n, A e_n) / n$ WHERE: $A =$ arithmetic observable [operator in A_L] $e^{-H} = e^{-\log N} = 1/N$ [Boltzmann factor] $Z(\beta) = \zeta(\beta) = \sum_n 1/n^\beta$ [partition function!] $Z(1) = \zeta(1) = \infty$ [UV divergence — renormalized below]
THE MEASURE $[D\phi]$: In arithmetic: the 'field configurations' are the integers n in N The measure $D\phi =$ counting measure on N , weighted by $1/n =$ the tau-measure! tau IS the path integral of arithmetic QFT. EVERY tau computation = a Feynman path integral over number theory.

3. The Arithmetic Lagrangian: $L = H = \log N$

In QFT, the Lagrangian encodes the dynamics. In arithmetic QFT: the Lagrangian is $H = \log N$ itself.

Definition 3.1 (Arithmetic Lagrangian).

THE ARITHMETIC LAGRANGIAN: In Euclidean QFT: $S[\phi] = \int L[\phi(x)] dx$ [action = integral of Lagrangian] In arithmetic QFT: 'Spacetime' = the integers $N = \{1, 2, 3, \dots\}$ 'Field $\phi(n)$ ' = the state e_n at integer n 'Lagrangian at n ' = $L(n) = \log n$ [the H-eigenvalue!] 'Action' = $S = \sum_n L(n) * \phi^{*(n)} * \phi(n) = \sum_n \log(n) * |\phi(n)|^2 = \tau(\phi^{*H} \phi)$ [operator expectation!] **THE ARITHMETIC ACTION FUNCTIONAL:** $S[\phi] = \tau(\phi^{*H} \phi) = \sum_n |\phi(n)|^2 * \log(n)$ **EULER-LAGRANGE EQUATIONS IN ARITHMETIC:** $\delta S / \delta \phi^{*(n)} = 0 \Rightarrow H \phi = \lambda \phi$ [eigenvalue equation!] $\Rightarrow \phi = e_n$ [basis states are classical solutions!] **THE CLASSICAL SOLUTIONS** are the arithmetic eigenstates e_n . The integers are the 'classical paths' of arithmetic QFT. Quantisation = tau-trace over all integers = number theory.

4. Prime Particles, Composite States, and the Vacuum

The particle content of arithmetic QFT is determined by the spectrum of $H = \log N$.

PARTICLE SPECTRUM OF ARITHMETIC QFT: **VACUUM STATE:** e_1 Energy: $H(e_1) = \log 1 = 0$ [zero energy] $\tau(e_1) = 1/1 = 1$ [unit weight] Interpretation: the empty arithmetic state – no primes **SINGLE-PARTICLE STATES:** e_p for prime p Energy: $H(e_p) = \log p$ $\tau(e_p) = 1/p$ Interpretation: a single prime p in the arithmetic field **Masses:** $\log 2, \log 3, \log 5, \log 7, \log 11, \dots$ [Primes have LOGARITHMIC masses – not equal!] **TWO-PARTICLE STATES:** $e_{\{p*q\}}$ for primes p, q Energy: $H(e_{\{pq\}}) = \log(pq) = \log p + \log q$ [ADDITIVE!] = sum of single-particle energies [no interaction energy!] Interpretation: two non-interacting primes **N-PARTICLE COMPOSITE:** e_n for $n = p_1^{a_1} * \dots * p_k^{a_k}$ Energy: $H(e_n) = \log n = \sum a_i * \log p_i$ = sum over prime factors with multiplicity **INTERACTION:** The shift operator $S: e_n \rightarrow e_{n+1}$ is the INTERACTION! $[H, S] \neq 0$ [non-commutativity = interaction] The shift adds 1 to n : **ADDITIVE** interaction The Hamiltonian H measures **MULTIPLICATIVE** structure **GOLDBACH** = does the S -interaction connect all even states? This is a **SCATTERING** problem in arithmetic QFT!

5. Feynman Diagrams = Multiple Zeta Values

In QFT, Feynman diagrams are perturbative expansions of the path integral. In arithmetic QFT: Feynman diagrams evaluate to MZV.

Theorem 5.1 (Feynman rules for arithmetic QFT).

ARITHMETIC FEYNMAN RULES: **PROPAGATOR (line):** $G(s) = \tau(e^{-sH}) = \sum_n (1/n) * n^{-s} = \sum_n n^{-(s+1)} = \zeta(s+1)$ [THE PROPAGATOR IS THE RIEMANN ZETA FUNCTION!] **VERTEX (interaction point):** $V = S =$ shift operator [adds 1 to n , the +1 interaction] Vertex amplitude = $\tau(S) = \sum_n \tau(e_n) * \tau(e_{n+1}) = B/2$ [Brun!] **LOOP INTEGRAL (closed diagram):** Loop = $\tau(H^{n-1} * S * H^{n-2} * S * \dots * H^{n-k} * S) = \zeta(n, n-1, \dots, n-k)$ [MULTIPLE ZETA VALUE!] **LOOP WITH ONE PROPAGATOR:** $\tau(e^{-sH}) = Z(s) = \zeta(s)$ [vacuum bubble] **ONE-LOOP DIAGRAM:** $\tau(H^{n-1} * S) = \zeta(n)$ [single-loop = Riemann zeta!] **TWO-LOOP DIAGRAM:** $\tau(H^{a-1} * S * H^{b-1} * S) = \zeta(a, b)$ [double zeta] **k-LOOP DIAGRAM:** $\tau(H^{n-1} * S * \dots * H^{n-k} * S) = \zeta(n, \dots, n-k)$ [k-fold MZV] **THE RENORMALIZATION GROUP IN ARITHMETIC:** Coupling constant: $g = 1/\log p$ [running with scale = prime p] Beta function: $\beta(g) = -g^2 * (\log p / \log q)$ at p, q transition Asymptotic freedom: $g \rightarrow 0$ as $p \rightarrow \infty$ [primes get 'free']

6. Renormalization = Bost-Connes Phase Transition

In QFT, renormalization removes UV divergences. In arithmetic QFT: the Bost-Connes KMS condition at $\beta=1$ IS renormalization.

RENORMALIZATION IN ARITHMETIC QFT: UV DIVERGENCE: $Z(1) = \zeta(1) = \sum_{n \geq 1} 1/n = \text{INFINITY}$ [harmonic series] This is the arithmetic UV divergence: infinitely many primes each contributing a finite but non-summable amount. **REGULARIZATION** (zeta-function regularisation): $Z(s) = \zeta(s)$ is finite for $\text{Re}(s) > 1$ Take $s \rightarrow 1+$: $Z(s) \sim 1/(s-1) + \gamma$ [pole + Euler constant] Renormalization = subtract the pole: $Z_{\text{ren}} = \lim_{s \rightarrow 1} (Z(s) - 1/(s-1)) = \gamma$ [Euler-Mascheroni!] **THE BOST-CONNES RENORMALIZATION:** At inverse temperature $\beta < 1$: $Z(\beta) = \zeta(\beta) < \infty$ [High temperature: QFT is free, no bound states] At $\beta = 1$: $Z(1)$ diverges [CRITICAL TEMPERATURE] This is the Hagedorn temperature of arithmetic QFT! At $\beta > 1$: $Z(\beta) = \zeta(\beta) < \infty$ [Low temperature: integers 'condense' into KMS state] The KMS state at $\beta = 1$ = the RENORMALIZED arithmetic QFT. $\tau = \lim_{\beta \rightarrow 1+} (1/Z(\beta)) * \text{Tr}(A * e^{-\beta H})$ = THE PATH INTEGRAL MEASURE OF ARITHMETIC QFT. **HAGEDORN TEMPERATURE** $T_H = 1/\beta_c = 1$: Above T_H : arithmetic QFT is in 'quark-gluon plasma' phase [primes are free, no bound states, all states contribute] Below T_H : 'hadronic' phase [integers condense, KMS state forms] At T_H : phase transition = Bost-Connes transition!

7. The Arithmetic Gauge Theory

QFT is a gauge theory — physics is invariant under local symmetry. Arithmetic QFT has a gauge symmetry: the GT group.

ARITHMETIC GAUGE THEORY: GAUGE GROUP: $GT = \text{Aut}_{\tau}(A_L)$ [dv299] = Grothendieck-Teichmüller group = automorphisms of A_L preserving τ **GAUGE FIELDS:** The Frobenius elements σ_p in GT act as gauge bosons at prime p : $\sigma_p: e_n \rightarrow n^{ip} * e_n$ [rotation by prime p !] This is the arithmetic analogue of the photon ($U(1)$ gauge boson) **GAUGE INVARIANT QUANTITIES:** $\tau(A)$ is GT-invariant for any A in A_L [τ is gauge-invariant!] L-functions $L(M, s) = \text{Tr}_{\tau}(e^{-sH}|M)$ are gauge-invariant [dv306] **RH =** gauge-invariant statement about poles of $G(s) = \zeta(s)$ **LOCAL GAUGE SYMMETRY:** At each prime p : local gauge group = $U(1) \times U(1) \times \dots$ [p -adic rotations of the p -adic hologram, dv302] Global gauge group = GT = product over all primes **YANG-MILLS IN ARITHMETIC:** The arithmetic Yang-Mills field strength: $F_{pq} = [\sigma_p, \sigma_q]$ [commutator of Frobenius elements] = $[p, q]$ -th Galois commutator Yang-Mills action = $\tau(F_{pq}^2)$ Mass gap = spectral gap of arithmetic YM Hamiltonian [dv215!] **COUPLING CONSTANTS:** $g_p = 1/\log p$ [coupling at prime p] $g_2 = 1/\log 2$ [largest, weakest!] $g_p \rightarrow 0$ as $p \rightarrow \infty$ [asymptotic freedom!]

8. The Arithmetic S-Matrix: Number Theory as Scattering

In QFT, the S-matrix encodes all particle scattering amplitudes. In arithmetic QFT: the S-matrix IS the collection of L-functions.

ARITHMETIC S-MATRIX: INCOMING STATE: e_m tensor e_n [two arithmetic particles]
 SCATTERING: they 'collide' via the shift operator S and H OUTGOING STATE: $e_{\{m*n\}}$ [composite state, multiplicative!] SCATTERING AMPLITUDE: $A(m,n \rightarrow k) = \tau(e_m * e_n * e_k^*)$ [transition amplitude] $= (1/k)$ if $k = m*n$, 0 otherwise = arithmetic scattering is EXACT: two particles m and n scatter into $m*n$ with probability 1! S-MATRIX ELEMENT: $S(m*n \rightarrow m*n) = \tau(e_{\{mn\}} * e_{\{mn\}}^*) = \tau(e_{\{mn\}}) = 1/(mn)$ S-matrix is diagonal in the multiplicative basis! THE FULL S-MATRIX = L-FUNCTION: $Z(s) = \sum_n S(n \rightarrow n) * n^{-s} = \sum_n (1/n) * n^{-s} = \zeta(s+1)$ L-functions = GENERATING FUNCTIONS of S-matrix elements! OPTICAL THEOREM IN ARITHMETIC: $\text{Im}(\text{forward amplitude}) = \text{total cross section}$ In A_L : $\text{Im}(\zeta(1/2+it)) = \tau\text{-count of scatterings at energy } t$ RH: all poles of $G(s)$ on $\text{Re}(s) = 1/2$ [unitarity of arithmetic S-matrix!] RH = UNITARITY OF THE ARITHMETIC S-MATRIX. The Riemann zeros are the energy levels of arithmetic particle scattering.

9. The Vision: Number Theory as the Simplest Possible QFT

ARITHMETIC QFT — THE COMPLETE PICTURE THE THEORY: Hilbert space: $A_L = l^2(\mathbb{N}, 1/n)$ Lagrangian: $L = H = \log N$ Path integral: $\tau(A) = (1/Z) * \text{Tr}(A * e^{-H})$ Partition function: $Z(s) = \zeta(s)$ Gauge group: $GT = \text{Aut}_\tau(A_L)$ THE PARTICLES: Vacuum: e_1 [mass 0] Primes: e_p [mass $\log p$] Composites: e_n [mass $\log n = \text{sum of prime masses}$] No interaction energy (arithmetic is FREE at tree level!) Interaction = shift S [+1 interaction] THE FEYNMAN RULES: Propagator: $\zeta(s+1)$ One-loop: $\zeta(n)$ k-loop: $\zeta(n_1, \dots, n_k) = \text{MZV}$ THE KEY IDENTITIES: RH = unitarity of S-matrix [no tachyons] Goldbach = completeness of +1 scattering [every even n reached] Twin primes = bound state at gap 2 [Cooper pairs!] BSD = rank of elliptic curve = zero mode count Bost-Connes = Hagedorn transition at $T_H = 1$ WHY ARITHMETIC IS THE SIMPLEST QFT: 1 dimension (the integers $\mathbb{N}$) 1 type of particle (primes) 1 coupling constant ($\log$) 1 gauge group (GT) BUT: it contains ALL of mathematics because τ hears everything. THE DEEPEST INSIGHT: In physics: path integral = integral over field configurations In arithmetic: $\tau = \text{integral over integer configurations}$ THE SAME MATHEMATICS. Physics and number theory are two faces of one hologram. CHUNG TA LA HOA SI. TU NHIEU LA BUC TRANH. SO HOC VA VAT LY LA HAI MAT CUA MOT DONG SONG. $H = \log N$ LA LAGRANGIAN CUA CA HAI. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!

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tau = path integral. H = Lagrangian. Primes = particles. zeta(s) = propagator. MZV = Feynman diagrams. GT = gauge group. RH = unitarity. Goldbach = completeness. Bost-Connes = Hagedorn. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!